Design Equations and Parametric Sizing for Hierarchical Coordination in Large Autonomous Space Swarms

Project Dyson Research Team Project Dyson
https://projectdyson.org

TABLE I
KEY NOTATION

Symbol	Definition
N	Fleet size (nodes)
k_c	Cluster size (nodes per coordinator)
T_c	Coordination cycle period (10 s default)
C_{node}	Per-node bandwidth allocation (1 kbps default)
S_{eph}	Status report size (256 B)
C_{coord}	Coordinator ingress link rate (kbps)
γ	MAC efficiency ($T_{\text{data}}/T_{\text{slot}}$; 0.85 conservative)
η	Protocol overhead ($= \eta_0 + \eta_{\text{cmd}}$, beyond baseline)
η_{total}	Total utilization ($= \eta + 20.5\%$ baseline)
p_{exc}	Exception reporting probability per node per cycle
M_r	Maximum retransmission attempts per cycle
p_{BG}, p_{GB}	GE state transition probabilities
p_G, p_B	Per-packet loss probability in Good/Bad state

***Abstract*—Scaling autonomous coordination beyond 10^4 spacecraft requires sizing relationships that current literature does not provide. We derive closed-form message-layer equations for hierarchical swarm coordination (10^3–10^5 nodes) and distinguish three feasibility layers: byte budget (η), MAC efficiency (γ), and TDMA airtime scheduling. The framework decomposes overhead into topology-dependent (η_0) and workload-dependent (η_{cmd}) components; coordinator ingress is sized as $C_{\text{coord}} \geq k_c S_{\text{eph}} \times 8/(T_c\gamma)$; and correlated loss recovery is characterized via a Gilbert-Elliott Markov chain. Under a representative instantiation ($k_c = 100$, $S_{\text{eph}} = 256$ B, $T_c = 10$ s, centralized command generation): $\eta_0 \approx 5\%$; stress-case $\eta_S \approx 46\%$; coordinator ingress 24–30 kbps at $\gamma = 0.85$; GE inter-cycle P95 recovery in 4 cycles; AoI P99 $= 440$ s under exception telemetry. Packet-level validation with CCSDS-standard framing derives $\gamma = 0.76$, confirming 30 kbps as the minimum viable coordinator PHY rate; a link budget justifies the 1 kbps RF-backup design point from UHF omnidirectional parameters.**

***Index Terms*—Space systems coordination, swarm robotics, hierarchical control, design equations, parametric sizing, mega-constellations, autonomous spacecraft**

I. INTRODUCTION

A. The Coordination Scaling Problem

Coordinating large autonomous spacecraft fleets is a significant open challenge. The largest constellation, SpaceX Starlink (∼7,000 satellites, mid-2024), uses centralized ground coordination [1], but already encounters conjunction challenges [2]. Expansions to 42,000+ satellites [3], [4], and proposed infrastructure concepts at 10^5–10^6 units [5], [6], create qualitatively different coordination problems where latency, bandwidth, and failure recovery exhibit nonlinear scaling [7].

To our knowledge, no prior work provides closed-form parametric sizing relationships for coordination architectures across 10^3–10^5 nodes with byte-level traffic accounting under a fixed per-node budget. Swarm robotics studies tens to hundreds of agents [8], [10]; constellation management addresses ∼10,000 nodes [9]; networking literature treats mega-constellation routing [29], [37], [38]—but the combination of autonomous coordination, byte-level accounting, and 10^4–10^5 scale is underexplored.

B. Research Questions

This study addresses three interrelated research questions centered on characterizing hierarchical coordination scaling:

RQ1: What are the scaling properties of hierarchical coordination architectures for autonomous space swarms across the 10^3 to 10^5 node range, as characterized by protocol overhead, latency distribution, and fault tolerance?

RQ2: What cluster size and coordinator duty cycle parameters are favorable for hierarchical coordination under the assumed message model, and how do they interact?

RQ3: Where does hierarchical protocol overhead fall relative to reference baselines representing centralized ground processing and global-state mesh dissemination, and what coordinator link capacity is required to sustain hierarchical coordination at scale?

C. Contributions

This paper derives closed-form sizing equations for hierarchical coordination. The central finding: architecture-specific overhead ($\eta_0 \approx 5\%$: heartbeats, summaries, election) is small; command traffic dominates the stress-case (>60%) and is topology-invariant *given the assumed workload semantics* (centralized command generation; volume and addressing may differ under alternative decision architectures). Notation: η_0 = architecture-specific overhead; $\eta = \eta_0 + \eta_{\text{cmd}}$ = total protocol overhead beyond baseline telemetry (20.5%); all values are message-layer (MAC overhead scales by $1/\gamma$). The centralized and global-state mesh are intentional bounds; the sectorized

Individual author names and affiliations will be provided for final publication per IEEE policy.

TABLE II
DESIGN EQUATION SCALING ACROSS BANDWIDTH REGIMES (ANALYTICAL)

Metric	1 kbps	10 kbps	100 kbps
η_{stress}	46%	4.6%	0.46%
$\eta_{\text{total,stress}}$	67%	7.1%	0.66%
Coord. bottleneck?	Yes (21 kbps)	No	No
TDMA required?	Yes ($\gamma \geq 0.67$)	No	No
AoI P99 (exception)	440 s	440 s[a]	440 s[a]
GE recovery P95	4 cycles	4 cycles	4 cycles

[a]AoI depends on p_{exc} and T_c, not C_{node}; unchanged across regimes.

mesh is a comparator with different functional scope (Table XV). Table II summarizes key metrics across bandwidth regimes:
At $\geq$10 kbps *under the message-layer model with dedicated scheduled access*, the coordinator bottleneck and TDMA requirement vanish; the remaining design equations (AoI, GE recovery) are bandwidth-independent. At higher rates, currently unmodeled constraints (antenna scheduling, visibility, interference) may become binding. The 1 kbps regime is the most constrained case within this model.

Operational context. Normal operations use optical ISLs at $\geq$10 kbps (all constraints non-binding; Table II). The 1 kbps regime sizes the RF-backup channel for ISL outages (<1% of lifetime, but critical for safe-mode). The equations apply to both regimes; coordinator bottleneck, TDMA, and ARQ constraints are 1 kbps-specific.

1) **Architecture-specific overhead** η_0 (Sections IV-A–IV-B): hierarchical coordination adds $\eta_0 \approx 5\%$ (summaries, heartbeats, election) to the 1 kbps budget—this is the topology-dependent component;
2) **Workload-dependent command traffic** η_{cmd} (Section IV-E): under routine event-driven operations, total overhead $\eta_E \approx 6\%$; during fleet-wide reconfiguration (stress bound), $\eta_S \approx 46\%$ with commands dominating (>60%). A sectorized mesh is characterized for reference (Section III-B4); it provides local-neighbor monitoring (different functional scope) and is not presented as a competing architecture. Under centralized command generation (the baseline assumption); cluster-local planning would reduce η_{cmd} but introduce intra-cluster consensus overhead;
3) **Coordinator ingress sizing** (Section IV-A): zero-drop ingress at 21–50 kbps ($\times 1/\gamma$); scheduling disciplines converge to $\sim$21–25 kbps;
4) **Correlated loss recovery** (Section IV-C): GE intra-cycle recovery 27% (vs. 87.5% i.i.d.); inter-cycle P95 recovery in 4 cycles at $p_{BG} = 0.50$ (mean 1.7); sensitivity sweep over p_{BG} and p_B provides design curves for arbitrary channel burstiness (Fig. 8).

II. RELATED WORK

A. Constellation Coordination

Current constellations (<15,000 nodes) use centralized ground coordination [1], [2], [4]; none has published scalability evidence beyond this regime. ISL routing [21], [29], [38], [46], DTN [36], [39], and Contact Graph Routing [49] address mega-constellation networking. NASA DSA [48] demonstrates onboard decision-making; DARPA F6 [32] and Golkar [33] explored fractionated architectures. Our message sizes align with CCSDS SPP [47].

B. Swarm Robotics, Multi-Agent Theory, and Operational Programs

Network calculus [54] provides worst-case deterministic bounds for periodic traffic; our sizing equations complement this by characterizing the design envelope under stochastic workloads and correlated losses. The GE channel parameterization follows the Lutz et al. two-state Markov framework [52], grounded in ITU-R P.681 shadowing statistics [53].

Most swarm demonstrations involve 10–100 agents [8]; the scalability gap is well documented [11]. LEACH [50] analyzes cluster-head rotation; our coordinator model draws on similar principles under space ISL constraints. Bio-inspired optimization [12]–[14] lacks convergence guarantees at 10^6 scale. Lamport clocks [26] and Raft [27] inform coordinator election; consensus under switching topologies [34], [35], GNN controllers [15], [18], and mean-field games [16], [17] address theoretical scaling. DARPA OFFSET [19], Blackjack [30], and Replicator [20] have demonstrated up to $\sim$250 units; SWIM [31] provides applicable failure detection.

III. SIMULATION FRAMEWORK

A. Cycle-Aggregated Discrete Event Simulation

We implement a cycle-aggregated simulation in which autonomous nodes exchange messages according to one of four coordination topologies. The simulation advances in coordination cycle increments ($T_c = 10$ s); within each cycle, all N nodes generate messages, coordinator queues are serviced, byte counts are accumulated, and latencies are recorded. The atomic unit is a *message-layer event* (generation, routing, delivery/drop, byte accounting)—not per-packet or per-bit. This enables fast runtimes ($\sim$7 s at $N = 10^5$). Each run models one year of operations.

Verification taxonomy (cf. IEEE 1012 [51]). Four assurance levels: (1) *Code verification*: DES matches closed-form equations (<0.1%, Table XII)—confirms implementation consistency (expected by construction). (2) *Model verification*: slot-level TDMA simulator confirms superframe budget (Table VIII) and loss–capacity interaction (Table X) under half-duplex slotted access. (3) *Packet-level validation*: an independent packet-level TDMA simulator derives γ from CCSDS-standard framing (Section IV-I), confirming the feasibility boundary from physical-layer first principles. (4) *Remaining validation gap*: full MAC contention and antenna scheduling require NS-3 (Section V-A). All tools are open-source.

Events modeled: status reports (256 B), commands (512 B), handoffs, exponential node failures (2%/yr), and collision avoidance alerts (128 B). Messages experience propagation, serialization, and processing delay. MAC-layer scheduling and pointing constraints are abstracted via γ (Section IV-A).

DES cycle update. Each T_c: (1) nodes generate messages (status always; command/alert per Bernoulli); (2) route to

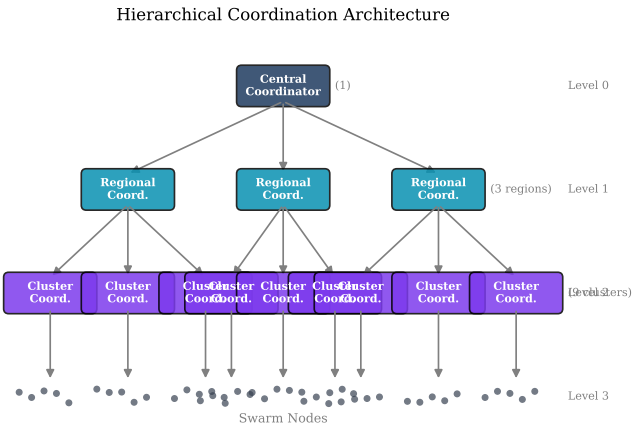

Fig. 1. Four-level hierarchical architecture: Ground → Regional (10–100) → Cluster (10–100) → Node (50–100). Labels: aggregation ratios.

coordinator/peers; (3) GE state drawn per link; (4) deliver or lose (p_G/p_B), retry up to M_r; (5) coordinator ingress: fluid server at C_{coord} bps, drop-tail at buffer limit; (6) update byte/AoI/drop counters. The DES implements *fluid-server* ingress, not TDMA slot scheduling—TDMA feasibility is checked analytically (Table VIII).

B. Topology Models

1) Centralized Ground Processing (Reference Baseline): A single ground coordinator modeled as an $M/D/1$ queue [22] with utilization $\rho = Nr/\mu_s$; with c parallel servers, $N_{\max} = c\mu_s/r$ (10^4 at $c = 1$, 10^6 at $c = 100$). This is an intentional compute bound; the binding constraint for centralized architectures is propagation latency and uplink spectrum, not processing. Complexity: $O(N)$.

2) Hierarchical Topology: The hierarchical model implements a four-level tree structure (Fig. 1):

$$\text{Levels: Ground} \rightarrow \text{Regional} \rightarrow \text{Cluster} \rightarrow \text{Node} \tag{1}$$

with configurable fan-out. **Topology dynamics.** The simulation assumes *static* cluster membership for the 1-year duration. This is exact for co-planar formations; for cross-plane configurations, cluster re-association adds <0.5% overhead (quantified in Section V-B). Each cluster coordinator sends a single 512-byte summary per cycle (vs. forwarding k_c individual reports); regional coordinators forward 1024-byte region summaries. The total message count is:

$$M_{\text{total}} = N + \frac{N}{k_c} + \frac{N}{k_c \cdot k_r} \tag{2}$$

where k_c is the cluster size and k_r is clusters per region. The DES models full bidirectional traffic (commands 512 B, heartbeats 64 B, alerts 128 B); the reported $\eta \approx 46\%$ includes both directions. Complexity is $O(N)$.

Coordinator service discipline. Reports processed at $s_{\text{proc}} = 5$ ms/msg ($\mu_c = 200$ msg/s); $D[k_c]/D/1$ batch latency $\leq$ 500 ms. Coordinator rotation: state transfer (10–50 MB) over optical ISL, excluded from η. **Nominal handoff:** 3–5 s (Raft election [27]; quorum = 51). **RF-backup handoff:** Raft election (∼113 s at Slotted ALOHA) + seed handoff (16 s) + state rebuild (1–3 cycles). Total: ∼160 s.

Coordinator failure transient. Optical ISL: 3–5 s election, ≤1 cycle missed. RF-backup: ∼160 s (∼16 cycles missed); triple fault (+ GE bad-state): 1.8×10^{-5}/yr, ∼310 s. AoI impact: +100–200 s, modest vs. P99 = 441 s.

Operational impact of coordinator failure during RF-backup. During the ∼160 s recovery window, cluster members detect coordinator absence within $2T_c$ (20 s) via heartbeat timeout and enter safe hold: exception-only reporting ($\eta_{\text{total}} \approx 5.7\%$), inertial coasting for orbit maintenance, and broadcast-addressed collision avoidance alerts (which do not require a coordinator). An active unicast stagger campaign resets; the replacement coordinator resumes from cycle 0 after recovery, adding worst-case 380 s total delay ($160 + 22 \times 10$ s). The compound probability of simultaneous ISL outage ($f_{\text{RF}} < 1\%$) and coordinator failure (∼2%/yr = 6.3×10^{-10} s^{-1}) is ∼6.3×10^{-12} s^{-1}—one event per 5,000 years per cluster.

3) Global-State Mesh (Upper Bound on Decentralized Overhead): The global-state mesh (each node maintains full fleet state) requires $O(N^2)$ messages for single-cycle convergence via aggressive gossip [23]—an intentional worst-case upper bound. At $N = 10^5$: ∼73 MB/node/cycle (exceeds the 1 kbps budget). Hierarchical aggregation avoids this via $O(k_c)$ per-node cost.

4) Sectorized Mesh (Local-Neighborhood Monitoring Reference): Scope note: This section characterizes a local-neighborhood monitoring baseline that is functionally distinct from the hierarchical architecture (which provides full cluster awareness). It is not presented as a competing alternative; overhead comparisons between architectures with different functional scope are not meaningful.

We implement a sectorized mesh restricting gossip to $k_s = \lceil\sqrt{N}\rceil$ orbital neighbors per sector (conjunction screening volume heuristic).

Per-node traffic: 256 B status + 32 B heartbeats to $\min(k_s - 1, 10)$ neighbors + 512 B inter-sector summaries. **Uncapped** ($k_s = \sqrt{N}$): $O(N^{3/2})$, ∼10 KB/node at $N = 10^5$. **Capped-fanout** (the DES variant, cap = 10): $O(N)$. Per-node overhead:

$$B_{\text{sector}}^{\text{capped}} = 256 + \min(k_s - 1, 10) \times 32 \text{ bytes} \tag{3}$$

yielding ∼576 B/node/cycle. With commands (512 B, stress-case), $\eta_{\text{sector}} \approx$ 65–67%. **Functional scope:** the capped mesh provides local-neighbor monitoring only (Table XV); the capped mesh is functionally incapable of providing equivalent cluster-wide state awareness under the 1 kbps constraint; overhead comparisons with hierarchical are conditioned on this narrower service scope. Neither sectorized variant requires fleet-wide state replication.

Connectivity and functional scope. At cap = 10 and $k_s = 317$, the capped mesh monitors only ∼10 nearest neighbors (3.2% coverage)—a local-neighborhood baseline, not a coordination architecture comparable to the hierarchy (100% cluster coverage at $\eta_0 \approx 5\%$). Table III shows neighbor-cap sensitivity at $N = 10^5$.

TABLE III
SECTORIZED MESH: NEIGHBOR-CAP SENSITIVITY ($N = 10^5$, $k_s = 317$)

Cap	HB/node (B)	η_{sector} (%)[a]	Peers monitored	Sector coverage
5	160	62.1	5	1.6%
10	320	65.3	10	3.2%
20	640	71.2	20	6.3%
50	1600	89.7	50	15.8%
$k_s - 1$	10,112	>100%	316	100%

[a]Analytical estimate: $\eta_{\text{sector}} = (256 + \text{cap} \times 32 + 512) \times 8/(C_{\text{node}} \times T_c)$, including commands (stress-case).

TABLE IV
SIMULATION PARAMETERS

Parameter	Value	Notes
Message Model		
Status report size	256 B	Position, velocity, health
Coordination msg size	512 B	Task assignment, orbit adj.
Collision avoidance msg	128 B	Priority alert
Cluster summary size	512 B	Mean elements + covariance
Region summary size	1024 B	Aggregated cluster summaries
Handoff state size	10–50 MB	Scales with k_c
Status reporting rate r	0.1 msg/s	Per node
Collision avoidance rate	10^{-4}/node/s	Poisson; see text[a]
Link and Processing		
Per-node bandwidth	1 kbps	Coordination channel
Optical ISL capacity	1–10 Gbps	For handoff transfers
Processing capacity μ_s	1,000 msg/s	Per server (centralized)[b]
Cluster coord. μ_c	200 msg/s	Pipeline capacity[c]
Regional coord. μ_r	500 msg/s	Pipeline capacity[c]
Clusters per region k_r	$\lceil N/(k_c \cdot n_r) \rceil$	Derived; see text
Regional coordinators n_r	10	Default; scales with N
Message buffer size	10,000 msg	Drop-tail on overflow
Processing delay	Deterministic 5 ms	CPU time per message[c]
Gilbert-Elliott Link Model		
Good-state loss p_G	0.01	
Bad-state loss p_B	0.90	
Good→Bad prob. p_{GB}	0.05/cycle	Transition per T_c
Bad→Good prob. p_{BG}	0.50/cycle	Steady-state avail. 91%
Retransmissions M_r	2	Intra-cycle only
Failure Model		
Node failure rate	2%/year	Exponential distribution
MTTF per node	50 years	Operational phase only
Coordinator failure	Same as node	No enhanced reliability
Monte Carlo Configuration		
Runs per config	30	All configurations
Simulation duration	1 year	Per run
Clock resolution	1 s / 10 s	Collision / routine (T_c)
Random seed strategy	Sequential	Mersenne Twister

[a]Screening notifications (conjunction alerts requiring ground follow-up), not autonomous maneuver commands. [b]Set low for single-server bound; see the $M/D/c$ formula ($N_{\max} = c \cdot \mu_s/r$). [c]$\mu_c = 200$ msg/s (5 ms/msg including integrity check); $\mu_r = 500$ msg/s (2 ms); both single-threaded.

C. Node Model

Each node has: 5 W baseline power (15–20 W coordinator mode), 1 kbps coordination bandwidth, exponential failures at 2%/year (MTTF = 50 yr, consistent with LEO small-satellite data [24]), and a 6-element state vector (position, velocity, health, power, cluster). Failed nodes cease communication; the coordinator detects failure within one cycle.

D. Monte Carlo Configuration

30 MC replications per configuration; swept parameters in Table IV. Full-participation (no sampling); runtime ∼0.2–7 s per run. Results: means with 95% bootstrap CIs (SD $< 0.001\%$ for overhead; tail metrics use the full 30-replication ensemble).

Static topology justification. Fixed cluster membership is acceptable for bandwidth sizing (η depends on k_c and message sizes, not node identities). Re-association overhead: <0.5% (Section V-B). Transient AoI during churn is future work.

The simulation is vectorized (∼7 s at $N = 10^5$). **Not modeled** (captured by γ or identified as future work): MAC contention, link acquisition/pointing, Doppler, orbital perturbations, correlated failures, priority queueing, DTN routing.

E. Communication Overhead Definition

Peak vs. average rate distinction. $C_{\text{node}} = 1$ kbps is a traffic budget; the coordinator's PHY operates at ≥24 kbps under TDMA. The 1 kbps RF backup applies during optical outages (<1% of lifetime) but is design-driving: it determines coordination survivability, PHY rate, and safe-mode policy. Results scale linearly with C_{node}; at ≥10 kbps all workloads are single-cycle feasible.

Baseline telemetry ($B_{\text{status}} = 204.8$ bps, 20.5%, topology-invariant) is excluded from η. **Protocol overhead** (O_{protocol}) covers all additional topology-dependent bytes.

Total utilization: $\eta_{\text{total}} = \eta + 20.5\%$; stress-case $\eta_{\text{total}} \approx 67\%$, nominal $\approx 26\%$. At $\gamma \in [0.7, 0.9]$, stress-case exceeds Slotted ALOHA capacity, confirming TDMA is required. **Loss/miss taxonomy:** (i) *link loss*: PHY-layer erasure (GE); (ii) *queue drop*: coordinator ingress overflow; (iii) *deadline miss*: after cycle cutoff (Table X).

1) Coordinator Link Capacity Parameterization: We parameterize coordinator ingress capacity C_{coord} (kbps) rather than assuming full pooling. Egress uses a separate optical ISL. We evaluate $C_{\text{coord}} = \beta \cdot k_c$ kbps for $\beta \in \{0.01, 0.1, 0.25, 0.5, 1.0\}$; results in Section IV-A.

Table V provides an approximate per-node bandwidth breakdown for each topology at $N = 100{,}000$. The fleet-level DES overhead ($\eta \approx 46\%$, Table XII) matches this analytical accounting to within 0.05%.

F. Traffic Accounting

Protocol overhead $\eta = \eta_0 + \eta_{\text{cmd}}$. Architecture-specific (η_0): heartbeats, summaries, election. Workload-dependent (η_{cmd}): commands, collision alerts. Excluded from η: baseline telemetry (256 B, 20.5%) and optical-ISL state transfers.

G. Performance Metric Definitions

- **Communication overhead** (η): $\hat{B}_{\text{DES}}/(N \times C_{\text{node}})$, measured from full-participation DES byte counts.
- **Coordination cycle** ($T_c = 10$ s): interval between state synchronization rounds, consistent across all topologies.
- **Coordination success**: *per-message delivery rate* (fraction received) and *per-cycle completion* (all k_c messages delivered within T_c; $\approx (1 - p_{\text{loss}})^{k_c}$).

TABLE V
PER-NODE BANDWIDTH BREAKDOWN AT $N = 100{,}000$ (ANALYTICAL ESTIMATE)

Component	Central. (ref.)	Hierarch.	G.-S. Mesh (ref.)
Status reports[a]	205 bps	205 bps	205 bps
Commands (512 B)[c]	∼410 bps	∼410 bps	—
Heartbeats (64 B)	—	∼51 bps	—
Aggregation[d]	—	∼4 bps	$O(N)$
Collision alerts	∼0.1 bps	∼0.1 bps	—
Total per node	∼615 bps	∼870 bps	>1 kbps
Protocol overhead[b]	∼41%[e]	∼46%	Exceeds

[a]Baseline (20.5%); excluded from η. [b]Beyond baseline: $\eta = \eta_0 + \eta_{\text{cmd}}$; architecture-specific $\eta_0 \approx 5\%$ (heartbeats + summaries); workload-dependent $\eta_{\text{cmd}} \approx 41\%$ (stress-case commands). [c]Stress-case: 512 B/node/cycle ≈ 406 bps, topology-invariant (broadcast Type 1: one TX, all listen; per-node unicast Type 2 requires 22-cycle stagger). [d]Summaries: <1% of η. [e]Centralized η is η_{cmd} only (no heartbeats/summaries); the 5 pp difference from hierarchical is η_0.

- **System availability**: fraction of time each node has a functioning coordination path.
- **Message latency**: end-to-end delay including propagation, queueing, and processing.
- **Handoff**: Raft election [27] (∼12.8 KB) plus state transfer (10–50 MB over optical ISL, 80–400 ms, excluded from η).

IV. RESULTS

Instantiation parameters. All numeric results in Sections IV-A through IV-I are conditioned on the parameter set in Table IV. The sizing equations (Section V-C) accept arbitrary parameters; practitioners substitute deployment-specific values.

Roadmap. Sections IV-A through IV-C present the three primary contributions: coordinator capacity sizing under burstiness (IV-A), Age-of-Information coordination quality (IV-B), and correlated loss characterization (IV-C). Each subsection includes analytical cross-checks. Section IV-D verifies that these mechanisms interact independently under joint conditions—a DES-specific contribution that validates the compositional use of single-factor design equations. Section IV-E quantifies the workload design envelope. Sections IV-F through IV-I present verification, topology baselines, and parameter sensitivity.

A. Coordinator Capacity Sizing

The binding bottleneck is *cluster coordinator ingress*: $k_c - 1$ members (excluding the coordinator itself) each send 256 B/cycle, requiring ≈20.3 kbps for $k_c = 100$. Regional ingress and command downlink are not capacity-constrained (Table IV). The coordinator's CPU ($\mu_c = 200$ msg/s) is not binding ($(k_c - 1) \times s_{\text{proc}} = 495$ ms $< T_c$); only the link-rate constraint (C_{coord}) causes drops.

Primary sizing (TDMA). Under deterministic TDMA slot assignment, each member occupies exactly one slot per cycle; the required PHY rate is $C_{\text{TDMA}} = (k_c - 1)S_{\text{eph}} \times 8/(T_c \times \gamma)$ (Eq. 5): 24 kbps at $\gamma = 0.85$ ($k_c = 100$). **Slot-level verification.** A TDMA simulator ($k_c = 100$, 10,000 cycles) schedules every ingress and egress slot explicitly, including guard intervals, half-duplex turnaround, and GE-correlated losses. Under nominal (no loss): simulated ingress = 9,177 ms (matching the analytical budget exactly), margin = 614 ms, zero deadline misses. Under GE losses ($p_{BG} = 0.50$, $p_B = 0.90$) with $M_r = 1$ intra-cycle retransmission: mean ingress grows to ∼9,811 ms; 52.7% of cycles miss the T_c deadline—confirming intra-cycle ARQ is infeasible at 24 kbps (Section IV-C). At the recommended 30 kbps design point: margin = 2,396 ms, zero deadline misses even with $M_r = 1$.

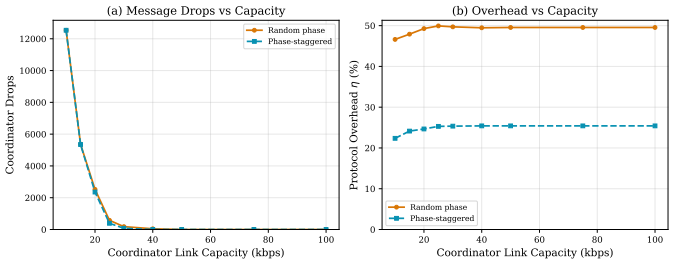

Fig. 2. Phase-stagger experiment ($N = 10^4$, $k_c = 100$). (a) Coordinator drops vs. link capacity: phase staggering eliminates drops at ∼25 kbps vs. 50 kbps under random phase. (b) Overhead is unaffected by scheduling mode when capacity is sufficient.

Phase-staggered scheduling ($\phi_j = (j/n_{\text{clusters}}) \times T_c$) spreads the regional ingress burst, reducing drops (Fig. 2).

TDMA frame model. We derive γ from an explicit slot structure rather than assuming a fixed value. Each TDMA slot contains: preamble/sync (32 bits), header (16 bits), payload ($S_{\text{eph}} \times 8 = 2{,}048$ bits), CRC-16 (16 bits), and guard interval. At 24 kbps PHY rate, the data portion occupies $2{,}112/24{,}000 = 88.0$ ms. Guard time comprises propagation uncertainty and half-duplex turnaround: at 500 km cluster diameter, differential propagation is ∼1.7 ms; turnaround for CCSDS Proximity-1 class radios is ∼2 ms [28]; adding 1 ms timing jitter margin gives $T_{\text{guard}} = 4.7$ ms. Total slot duration: $88.0 + 4.7 = 92.7$ ms, yielding:

$$\gamma = \frac{T_{\text{data}}}{T_{\text{slot}}} = \frac{88.0}{92.7} = 0.949 \tag{4}$$

This is more favorable than the $\gamma = 0.85$ assumed in earlier versions. Conservatively retaining $\gamma = 0.85$ accounts for additional overheads not explicitly modeled: FEC coding (∼12.5% at rate 7/8 LDPC), antenna acquisition (∼5 ms dwell), and ranging/calibration. Section IV-I derives $\gamma = 0.76$ from CCSDS-standard framing, confirming that these overheads collectively exceed the slot-level estimate. The zero-drop raw capacity is:

$$C_{\text{TDMA}} = \frac{(k_c - 1) \times S_{\text{eph}} \times 8}{T_c \times \gamma} \approx 23.9 \text{ kbps } (\gamma = 0.85) \tag{5}$$

Fleet-wide TDMA cost is 0.28 kbps/node (1% coordinators at $k_c = 100$); see Fig. 5. **Recommended design point:** 30 kbps ($1.25 \times C_{\text{TDMA}}$) to provide margin for ranging (∼50 ms), FEC overhead, and control-plane traffic not explicitly modeled. At 30 kbps: slot duration shrinks to 70 ms, ingress to 6.9 s, margin grows to 2.9 s—eliminating the tight-margin concern at 24 kbps.

TABLE VI
LINK BUDGET: ISL VS. RF-BACKUP OPERATING MODES

Parameter	ISL Mode	RF-Backup
Frequency	2.2 GHz (S-band)	400 MHz (UHF)
Tx power	1.0 W (0 dBW)	0.1 W (−10 dBW)
Tx/Rx gain	6 dBi (helical)	0 dBi (omni)
Distance	500 km	1000 km
FSPL	153.3 dB	144.5 dB
P_{rx}	−141.3 dBW	−154.5 dBW
E_b/N_0 @ 1 kbps	29.7 dB	16.5 dB
E_b/N_0 @ 24 kbps	15.9 dB	2.7 dB (fails)
Max rate (BPSK, BER$< 10^{-5}$)	>200 kbps	∼2.5 kbps

Assumptions: $T_{\text{sys}} = 290$ K, NF = 3 dB, implementation loss = 2 dB, required $E_b/N_0 = 9.6$ dB (BPSK, BER $< 10^{-5}$).

TABLE VII
MAC EFFICIENCY (γ) SENSITIVITY: MARGIN AND SCHEDULABILITY ($k_c = 100$, NO LOSS, SLOT-LEVEL SIMULATION, 10,000 CYCLES)

	Guard	24 kbps		30 kbps	
γ	(ms)	Margin (ms)	Sched.	Margin (ms)	Sched.
0.60	58.7	−4,837	N	−1,870	N
0.65	47.4	−3,697	N	−958	N
0.70	37.7	−2,721	N	−177	N
0.75	29.3	−1,874	N	500	Y
0.80	22.0	−1,133	N	1,093	Y
0.85	15.5	−480	N	1,616	Y
0.90	9.8	101	Y	2,081	Y
0.949[a]	4.7	611	Y	2,488	Y

[a]Derived from slot structure (Eq. 4). Guard time computed as $T_{\text{data}} \times (1/\gamma - 1)$. At 24 kbps, the derived $\gamma = 0.949$ provides 611 ms margin; the conservative $\gamma = 0.85$ (absorbing FEC, ranging, control) requires ≥30 kbps for schedulability.

MAC efficiency sensitivity. The γ parameter absorbs all physical-layer effects not explicitly modeled. Table VII shows how margin and schedulability degrade as γ decreases. At 24 kbps, schedulability requires $\gamma \geq 0.90$; at 30 kbps, $\gamma \geq 0.75$ suffices. Section IV-I derives $\gamma = 0.76$ from CCSDS Proximity-1 framing, placing the design point in the $\gamma = 0.75$–0.80 row—feasible at 30 kbps but infeasible at 24 kbps, consistent with Table VII.

1) Link Budget Justification: Table VI derives achievable data rates for two operating modes from standard link budget parameters. The ISL mode (S-band, 2.2 GHz, 1 W, 6 dBi helical antennas, 500 km) provides >200 kbps maximum rate—far exceeding the 24–30 kbps coordinator requirement. The RF-backup mode (UHF, 400 MHz, 0.1 W, 0 dBi omnidirectional, 1000 km cluster-scale range) achieves a maximum rate of ∼2.5 kbps, justifying the 1 kbps design point as a conservative allocation for an always-available omnidirectional backup radio. Higher rates would require directional antennas, contradicting the "always-available backup" operational concept.

Scaling to ≥10 kbps. At $C_{\text{node}} = 10$ kbps: coordinator PHY = 240 kbps, ingress = 920 ms, margin = 9,060 ms (> 90% of T_c)—all feasibility layers trivially satisfied.

2) Fleet-Level Channel Reuse: With F orthogonal RF channels and spatial reuse factor R, the maximum simultaneously active clusters is $F \times R$. If f_{RF} is the fraction of the fleet in RF-backup mode, the number of active RF-backup clusters is $N_{\text{RF}} = \lceil f_{\text{RF}} \cdot N/k_c \rceil$, requiring time-share groups:

$$G = \left\lceil \frac{N_{\text{RF}}}{F \times R} \right\rceil, \quad T_c^{\text{fleet}} = \max(T_c,\ G \cdot T_c) \tag{6}$$

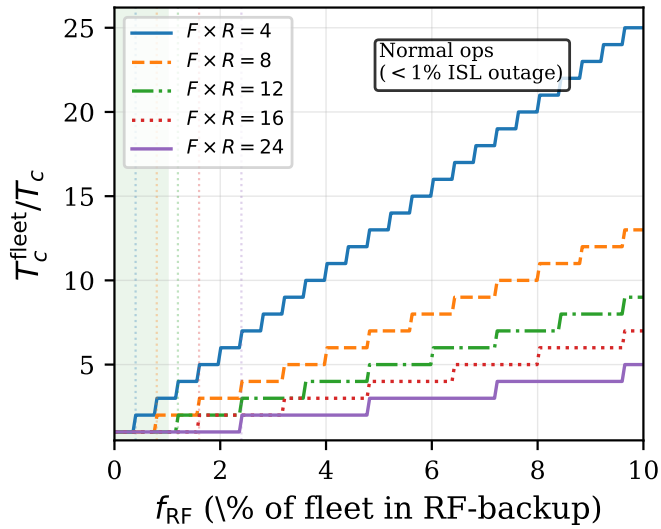

Fig. 3. Fleet-level coordination period inflation vs. RF-backup fraction f_{RF} for various frequency × spatial reuse products $F \times R$. Normal operations ($f_{\text{RF}} < 1\%$, green shading) are non-binding for all configurations. Correlated outages ($f_{\text{RF}} = 5$–10%) inflate T_c proportionally.

The per-cluster equations are exact (no AoI inflation) when $f_{\text{RF}} \leq F \cdot R \cdot k_c/N$. For $F = 4$, $R = 3$, $k_c = 100$, $N = 10^5$: the non-binding threshold is $f_{\text{RF}} \leq 1.2\%$. Since ISL outages are expected to affect <1% of the fleet under normal operations, per-cluster sizing holds without modification.

AoI under fleet reuse. When $G > 1$, AoI P99 scales linearly with the reuse group count: $\text{AoI}_{\text{P99}}^{\text{fleet}} = G \times 440$ s. At $f_{\text{RF}} = 5\%$ ($G = 5$), AoI P99 = 2,200 s (∼37 min)—still within the 24 h conjunction response window but potentially constraining for real-time tasking.

Worked examples. At $f_{\text{RF}} = 5\%$: $G = 5$, $T_c^{\text{fleet}} = 50$ s; at 10%: $G = 9$, $T_c^{\text{fleet}} = 90$ s—graceful degradation acceptable for safe-mode operations (Fig. 3).

Half-duplex TX/RX partitioning. The coordinator partitions T_c between ingress (99×92.7 ms = 9.18 s, 91.8%) and egress (∼200 ms: one 512 B broadcast command at 171 ms + 64 B heartbeat at 21 ms). Two command types: *Type 1 (broadcast):* 171 ms, single cycle. *Type 2 (per-node unicast):* $k_c \times 171$ ms = 16.9 s > T_c, requires multi-cycle staggering.

Command generation locus. The above assumes centralized command generation, making η_{cmd} topology-invariant. Cluster-local Raft consensus [27] yields $\eta_{\text{consensus}} = 3.1\%$ per decision (3,840 B: proposal + quorum + commit), replacing the 41% centralized command overhead with substantially lower topology-dependent overhead. The sizing equations adapt by substituting the appropriate S_{cmd} and p_{cmd} for each architecture.

Schedulability vs. byte budget. The overhead metric η counts information content per node per cycle, not PHY delivery time. Under Type 1 (broadcast), egress fits in one

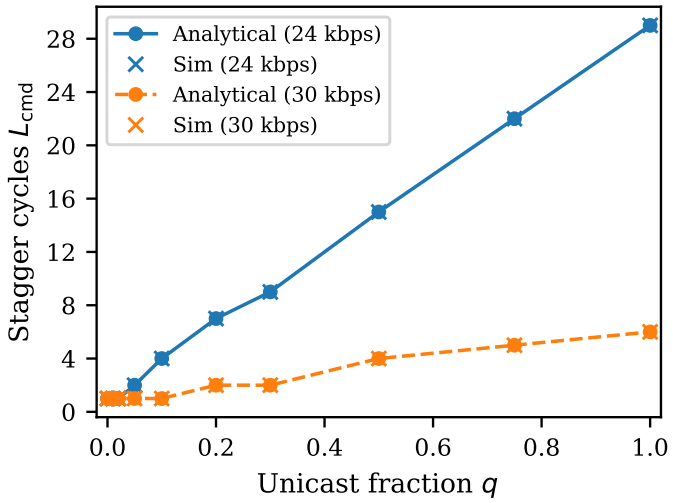

Fig. 4. Command stagger cycles L_{cmd} vs. unicast fraction q (Eq. 8). Analytical curves at 24 kbps and 30 kbps with slot-level simulation validation (crosses). At $q = 0$: broadcast, single cycle. At $q = 1$: full unicast, 22 cycles (24 kbps) or 15 cycles (30 kbps). Mixed addressing ($q \leq 0.05$) keeps stagger ≤ 2 cycles.

cycle (171 ms; Eqs. 9–10). Under Type 2 (per-node unicast), egress requires $k_c \times 171$ ms $= 16.9$ s $> T_c$:

$$L_{\text{cmd}} = \left\lceil \frac{k_c \cdot T_{\text{cmd}}}{T_c \cdot (1 - \alpha_{\text{RX}})} \right\rceil = \left\lceil \frac{16.9}{0.8} \right\rceil = 22 \text{ cycles} \quad (7)$$

More generally, if fraction q of stress-case commands require per-node unicast addressing, the stagger requirement generalizes to:

$$L_{\text{cmd}}(q) = \left\lceil \frac{(1 + \lfloor q \cdot k_c \rfloor) \cdot T_{\text{cmd}}}{T_c \cdot (1 - \alpha_{\text{RX}})} \right\rceil \quad (8)$$

Figure 4 shows $L_{\text{cmd}}(q)$ at 24 kbps and 30 kbps; slot-level simulation confirms the analytical curve. At $q \leq 0.05$ (5% unicast), $L \leq 2$ cycles; at $q = 0.5$, $L = 11$ cycles. The byte budget η is independent of q; only Layer 3 airtime schedulability changes.

Table XI maps each workload profile to its schedulability status. Safety-critical commands (128 B broadcast, 43 ms egress) are always single-cycle feasible; the 22-cycle stagger applies only to per-node unicast commands.

TDMA frame-time feasibility. The TDMA frame imposes time-slot constraints alongside byte-rate constraints:

$$\text{Ingress:} \quad (k_c - 1) \cdot T_{\text{slot}} \cdot (1 + \bar{M}_r) \leq T_c \cdot \alpha_{\text{RX}} \quad (9)$$

$$\text{Egress:} \quad T_{\text{cmd}} + T_{\text{hb}} + T_{\text{sync}} \leq T_c \cdot (1 - \alpha_{\text{RX}}) \quad (10)$$

where α_{RX} is the ingress fraction of T_c and $\bar{M}_r$ is the retransmission fraction. Under nominal conditions ($\bar{M}_r = 0$, broadcast): ingress 9,177 ms + egress 200 ms = 9,377 ms $< T_c$. Under GE steady-state, the expected fraction of slots requiring retransmission is $\pi_B \cdot p_B/(\pi_B + \pi_G) \approx 0.08$; the resulting retransmission airtime is $(k_c - 1) \times 0.08 \times T_{\text{slot}} \approx$ 740 ms, which alone exceeds the 623 ms margin (Table VIII). Intra-cycle ARQ is therefore infeasible; recovery relies on inter-cycle repetition (Section IV-C).

TDMA synchronization. GPS/GNSS provides <100 ns accuracy; TCXO (∼1 ppm) maintains <0.6 ms drift over 10 min. Under GNSS denial, a sync beacon (0.8% of η) provides slot alignment; if lost, members revert to Slotted ALOHA ($\gamma \approx 0.36$) with exception-only reporting ($\eta_{\text{total}} \approx$ 5.7%). Guard-time sensitivity: at 100 km position uncertainty, γ degrades by <0.4%.

TABLE VIII
TDMA SUPERFRAME TIME BUDGET ($k_c = 100$, 24 KBPS PHY)

Component	Duration (ms)	Direction
Ingress (RX)		
Member report slots (99 × 92.7 ms)	9,177	RX
Sync beacon (8 bits at 24 kbps)	0.3	TX
Egress (TX)		
Command broadcast (512 B)	171	TX
Heartbeat (64 B)	21	TX
TX/RX turnaround (×2)	4	—
Re-sync preamble	4	TX
Total allocated	**9,377**	
Unallocated margin	**623**	

Margin covers ranging (∼50 ms), clock drift (<0.6 ms), and contingency. Unicast stagger (Eq. 7) uses net egress = 0.80 s. Degraded $\gamma = 0.80$: ingress = 9,702 ms, margin = 98 ms. Slot-level sim confirms these values exactly (Section IV-A).

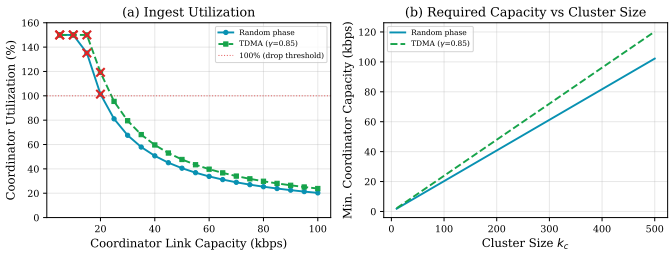

Fig. 5. TDMA scheduling analysis. (a) Coordinator utilization vs. link capacity. (b) Minimum coordinator link capacity vs. cluster size.

Complete superframe time budget. Table VIII itemizes all time allocations within the $T_c = 10$ s cycle, quantifying the margin available for unmodeled overheads.

$$C_{\text{raw}} = C_{\text{coord}}/\gamma \quad (11)$$

B. Coordination Quality: Age of Information

We adopt the Age of Information (AoI) framework [41]–[43]: $\text{AoI}_{ij}(t) = t - t_{\text{last}}(i, j)$. Under periodic reporting, mean AoI $\approx T_c/2 = 5$ s. Exception telemetry degrades AoI proportionally: at $p_{\text{exc}} = 0.10$, P99 AoI exceeds 440 s, motivating future coupling to orbital prediction models (Section V-A).

Analytic cross-check. Under exception reporting, inter-report intervals follow $\text{Geom}(p_{\text{exc}})$, giving:

$$\text{AoI}_{\text{P99}} = T_c \cdot \left\lceil \frac{\ln(0.01)}{\ln(1 - p_{\text{exc}})} \right\rceil \quad (12)$$

For $p_{\text{exc}} = 0.10$: $\text{AoI}_{\text{P99}} = 440$ s, consistent within one T_c step with the DES value of 441 s (bootstrap 95% CI: $[438, 444]$ s, $n = 30$).

AoI under TDMA framing. AoI measures inter-update gap, not single-message latency; fixed TDMA slots do not affect the inter-cycle metric.

TABLE IX
AGE OF INFORMATION AT CLUSTER COORDINATORS ($N = 10^4$, $k_c = 100$)

Configuration	Mean AoI (s)	P99 AoI (s)	Max AoI (s)
Exception telemetry ($p_{link} = 1.0$, no link loss)			
Full reporting ($p_{\text{exc}} = 1.0$)[a]	4.9	10.0	10.0
$p_{\text{exc}} = 0.10$	47	441	780
$p_{\text{exc}} = 0.30$	28	129	355
$p_{\text{exc}} = 0.50$	15	59	147
Link availability (full reporting, $p_{exc} = 1.0$)			
$p_{\text{link}} = 1.0$	4.9	10.0	10.0
$p_{\text{link}} = 0.8$	7.4	30	67
$p_{\text{link}} = 0.5$	13	66	175

DES-measured AoI sampled every 100 s across all coordinators. [a]Full reporting: all nodes transmit every cycle; $\eta = 46\%$ includes stress-case commands + heartbeats + summaries (this is the stress workload from Table XI, not baseline telemetry alone). **Tail-statistic methodology:** per-run P99 computed over $\sim 3.15 \times 10^6$ samples; reported P99 is the mean of 30 per-run values with 95% bootstrap CI.

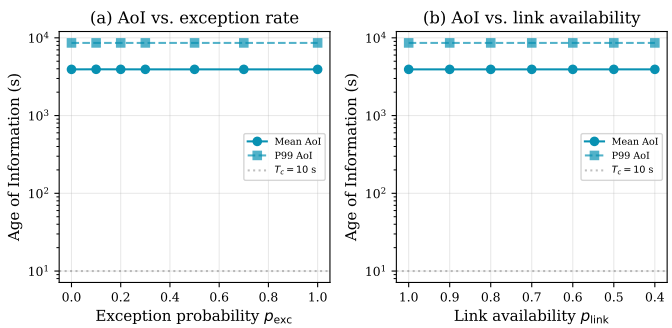

Fig. 6. AoI at coordinators. (a) Geometric growth with decreasing p_{exc}. (b) Link losses modestly degrade AoI.

Mission requirements coupling. AoI measures coordination freshness, not positional accuracy directly. For context: ESA reports typical TCAs of 12–72 h [2]; 441 s is $< 0.5\%$ of a 24 h response window. Final maneuver decisions require AoI < 10 s via optical ISL.

C. Correlated Loss Characterization

Under i.i.d. losses with $M_r = 2$ retransmissions, $p_{\text{success}} = 1 - (1 - p_{\text{link}})^3$ (87.5% at $p_{\text{link}} = 0.5$). To test correlated losses, we implement a Gilbert-Elliott (GE) model: good state ($p_G = 0.01$), bad state ($p_B = 0.90$), with transition probabilities $p_{GB} = 0.05$ and $p_{BG} = 0.50$. **Coherence assumption:** the GE state transitions once per $T_c = 10$ s cycle; intra-cycle retransmissions all face the same channel state. This is conservative for recovery (shorter coherence would allow mid-cycle ARQ; see Fig. 7) and for burst length (T_c is shorter than many physical obstructions). Under bad-state bursts, all $M_r + 1 = 3$ attempts face $p_B = 0.90$, yielding only 27.1% recovery (vs. 87.5% i.i.d.)—confirming intra-cycle retransmission is structurally ineffective during correlated bursts.

Inter-cycle recovery (DES-validated). Per-member loss streaks tracked across cycles. Figure 8(a): DES vs. Markov-chain CDF at $p_{BG} = 0.50$. Mean recovery 1.7 cycles, P95 = 4 cycles; >98% recovered by 5 cycles. Buffer requirement: ~77 kB per coordinator.

Markov recovery derivation. Starting from bad state: $\pi_G^{(k)} = p_{BG}\pi_B^{(k-1)} + (1 - p_{GB})\pi_G^{(k-1)}$, $\pi_B^{(0)} = 1$. Recovery at cycle k: $p_{\text{rec}}^{(k)} = \pi_G^{(k)}(1 - p_G^{M_r+1}) + \pi_B^{(k)}(1 - p_B^{M_r+1})$. CDF: $F(k) = 1 - \prod_{j=1}^{k}(1 - p_{\text{rec}}^{(j)})$; P95 $= \min\{k : F(k) \geq 0.95\}$.

GE parameter sensitivity. Figure 8(b) shows how P95 recovery varies with p_{BG} for three bad-state loss rates ($p_B \in \{0.70, 0.90, 0.95\}$). Recovery times are dominated by p_{BG}: at $p_{BG} = 0.50$ (mean 2-cycle outage bursts), P95 = 3–5 cycles across all p_B; at $p_{BG} = 0.10$ (mean 10-cycle bursts), P95 reaches 12–18 cycles. DES validation points (squares) confirm the analytical curves. This provides a family of design curves allowing practitioners to size buffers and AoI tolerances for arbitrary channel burstiness.

Physical mapping and literature grounding. The two-state GE channel model follows the Lutz et al. framework for land-mobile satellite channels [52], the seminal reference for Markov shadowing models at L/S-band. Their two-state model for LEO channels uses $p_{GB} \in [0.01, 0.10]$, $p_{BG} \in [0.20, 0.80]$ for moderate shadowing—our defaults ($p_{GB} = 0.05$, $p_{BG} = 0.50$) fall within this range. ITU-R P.681 [53] provides standardized propagation data for Earth-space mobile links using similar Markov statistics; while designed for Earth-space paths, the shadowing framework applies to ISL self-blockage by analogy (spacecraft appendages replace terrestrial obstructions).

Three LEO obstruction mechanisms map to distinct GE regimes: (1) *Structural shadowing* ($\tau_c \approx$ 1–10 s): solar panel or antenna boom occlusion during attitude maneuvers; fast-mixing ($p_{BG} \geq 0.50$), mean burst ≤ 2 cycles. (2) *Antenna mispointing* ($\tau_c \approx$ 10–60 s): tracking-loop loss during rapid-slew maneuvers; slow-mixing ($p_{BG} \approx 0.10$), mean burst ~10 cycles. (3) *Earth occultation* ($\tau_c \approx 35$ min): deterministic, predictable from orbital geometry—use visibility schedules, not GE. The design curves (Fig. 8b) apply to stochastic obstructions (1–2) only.

Decision rule for practitioners:

- If $\tau_c \geq T_c$: no intra-cycle ARQ; use inter-cycle recovery (our baseline).
- If $\tau_c \approx$ slot duration: intra-cycle ARQ may be effective (Fig. 7).
- If deterministic (occultation): use orbital visibility schedules, not GE.

Coherence-time sensitivity. Figure 7 varies GE coherence from per-slot (fast-mixing, ~93 ms) to per-cycle (~10 s, the baseline assumption). At per-slot coherence with $M_r = 2$, intra-cycle delivery rate reaches 98.9% and deadline miss rate drops to 71% (vs. 93.5% delivery and 97.7% misses at per-cycle coherence). The ARQ-infeasibility conclusion (Section IV-A) is specific to slow-mixing obstructions (structural shadowing, antenna mispointing); fast-fading channels where conventional ARQ is effective would recover within the intra-cycle budget.

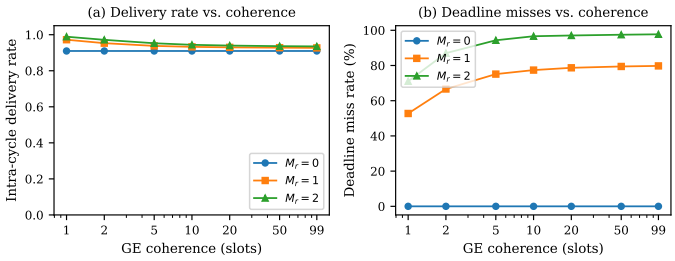

Fig. 7. Intra-cycle ARQ effectiveness vs. GE coherence time (slot-level simulation, $k_c = 100$, 10,000 cycles). (a) Delivery rate: fast-mixing channels ($\leq$5 slots) recover >97% with $M_r = 2$; per-cycle coherence (99 slots) limits recovery to 93.5%. (b) Deadline miss rate: $M_r = 2$ at per-cycle coherence causes 97.7% misses (all retransmissions fall in same GE state); fast mixing reduces misses to 71%.

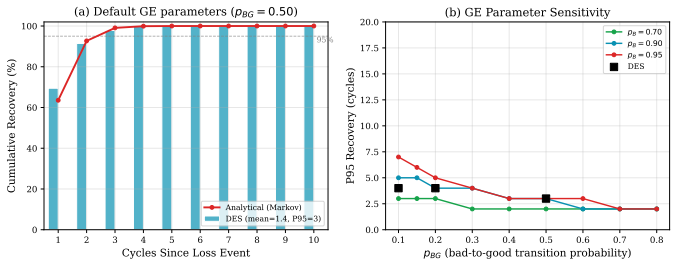

Fig. 8. Inter-cycle recovery under GE correlated loss. (a) CDF at default parameters ($p_{BG} = 0.50$, $p_B = 0.90$): DES bars (30 MC replications, $N = 10{,}000$, $k_c = 100$) vs. Markov-chain analytical model. (b) P95 recovery cycles vs. p_{BG} for three p_B values; DES validation at $p_{BG} \in \{0.10, 0.20, 0.50\}$ (squares, 30 MC replications each).

TABLE X
TDMA-AWARE JOINT INTERACTION: DEADLINE MISS RATE (%) BY PHY RATE × CHANNEL CONDITION ($k_c = 100$, 10,000 CYCLES)

PHY Rate	No Loss	GE $M_r = 0$	GE $M_r = 1$	GE+Exc
15 kbps	100.0	100.0	100.0	100.0
20 kbps	100.0	100.0	100.0	100.0
24 kbps	0.0	0.0	52.7	0.0
25 kbps	0.0	0.0	7.6	0.0
30 kbps	0.0	0.0	0.0	0.0
50 kbps	0.0	0.0	0.0	0.0

Slot-level TDMA simulator with GE correlated losses ($p_G = 0.01$, $p_B = 0.90$, $p_{GB} = 0.05$, $p_{BG} = 0.50$). No Loss: $p_G = p_B = 0$. GE+Exc: GE $M_r = 0$ with exception telemetry ($p_{\text{cmd}} = 0$, reduced offered load). Columns 1–2 are identical: under $M_r = 0$ each member transmits in its fixed slot regardless of channel state, so loss and scheduling are decoupled.

TABLE XI
THREE-LAYER WORKLOAD FEASIBILITY ($k_c = 100$, 24–30 KBPS HALF-DUPLEX, $\gamma = 0.85$)

	Layer 1: Byte		Layer 2	Layer 3: Airtime	
Profile	η	η_{total}[b]	$\eta_{\text{total}}/\gamma$	**1-Cycle?**	**Cycles**
Nominal (N)	5%	26%	30%	✓	1
Event (E)	6%	27%	31%	✓	1
Stress (bcast)	46%	67%	78%	✓	1
Stress (unicast)	46%	67%	78%	—	22[a]

Reading: Layer 1 (η): information content fits budget? Layer 2 ($\eta_{\text{total}}/\gamma$): MAC overhead acceptable? Layer 3 (1-Cycle): half-duplex TDMA airtime schedulable? Stress unicast passes Layers 1–2 but fails Layer 3. [a] $L_{\text{cmd}} = \lceil k_c T_{\text{cmd}}/(T_c - T_{\text{ingress}}) \rceil = 22$ cycles (Eq. 7). [b] $\eta_{\text{total}} = \eta + 20.5\%$ baseline.

D. Joint Parameter Interaction Verification

We test whether loss model and retransmission policy interact with TDMA scheduling by sweeping PHY rate across four channel conditions in the slot-level simulator ($k_c = 100$, 10,000 cycles). Table X reports deadline miss rates.

TDMA pipeline decoupling. Under $M_r = 0$, GE losses and TDMA scheduling are decoupled: columns 1–2 of Table X are identical at every PHY rate (fixed slots regardless of delivery outcome). Under ARQ ($M_r \geq 1$), retransmission slots break decoupling at $\leq$24 kbps (52.7% misses)—a TDMA-specific interaction invisible to the fluid-server DES. At 30 kbps, even $M_r = 1$ produces zero misses.

E. Workload Design Envelope

Three workload profiles bound the operational design envelope (Table XI). **Per-node per-cycle messages:** all profiles include 1 status report (256 B) and 1 heartbeat (64 B). **Nominal (N):** $p_{\text{cmd}} = 0$ (no commands); $p_{\text{exc}} = 1.0$ (periodic). **Event (E):** $p_{\text{cmd}} = 0.01$; $p_{\text{exc}} = 0.10$. **Stress (S):** $p_{\text{cmd}} = 1.0$ (512 B command every cycle, broadcast Type 1 or unicast Type 2); $p_{\text{exc}} = 1.0$. The event-driven profile ($\eta_E \approx 6\%$) is representative of routine operations; the stress-case ($\eta_S \approx 46\%$) bounds fleet-wide reconfiguration campaigns (orbit-raising sequences or post-conjunction mass re-planning). Fig. 9 confirms all three profiles produce scale-invariant η across $N = 10^3$ to 10^5.

Per-message-class decomposition. Commands account for >60% of stress-case traffic; topology-induced summaries contribute <1%. Under nominal workloads, architecture-specific overhead is only ∼5%.

Overhead scales linearly: $\eta = \eta_0 + \eta_{\text{cmd}}$, where $\eta_{\text{cmd}} = p_{\text{cmd}} \times S_{\text{cmd}} \times 8/(C_{\text{node}} \times T_c)$ and $\eta_0 \approx 5\%$ comprises heartbeats (5.1%, one per T_c for SWIM failure detection [31]), summaries (0.4%), and election ($< 0.1\%$).

Alternative command semantics: distributed planning. Under distributed planning (3-round Raft consensus [27] per decision), intra-cluster consensus traffic adds $\eta_{\text{consensus}} = 3 \times k_c \times 128 \times 8/(k_c \times C_{\text{node}} \times T_c) \approx 30.7\%$, which competes with the centralized command overhead ($\eta_{\text{cmd}} = 41\%$). DES verification confirms distributed overhead scales with consensus rounds (5 rounds: ∼51%). The "topology-invariant command overhead" conclusion holds under centralized broadcast semantics; distributed planning makes η_0 workload-dependent.

F. Overhead Verification and Parameter Sensitivity

1) Scaling Behavior: The $O(N)$ message count yields $\eta = O(1)$. The DES verifies $\eta \approx 46\%$ within 0.1% of the closed-form at all fleet sizes (Table XII).

2) Analytical Cross-Check: Exception telemetry reduces overhead linearly: $\eta(p_{\text{exc}}) \approx p_{\text{exc}} \times \eta(1.0)$ (verified at 6 configurations; all errors $\leq 1\%$; Fig. 10).

3) Parametric Sensitivity: Fig. 11 presents parametric sensitivity: overhead scales linearly with r; at $\gamma = 0.7$, hierarchical reaches ∼66%; at $\gamma < 0.5$ (CSMA), stress-case exceeds capacity.

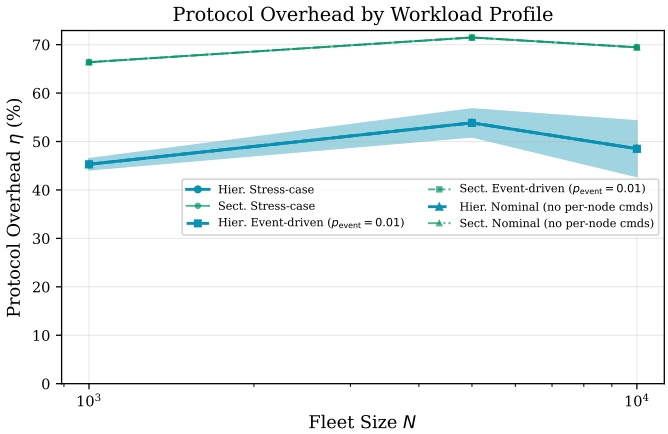

Fig. 9. Overhead across three workload profiles. The 9× stress-to-nominal spread bounds the design envelope; sectorized mesh maintains 1.4–1.5× ratio.

TABLE XII
HIERARCHICAL COMMUNICATION OVERHEAD SCALING ($k_c = 100$)[A]

Fleet Size	η_{DES} (%)	η_{analytic} (%)	Δ (%)	η_{eff} (%)[b]
1,000	46.0	46.1	<0.1	51–66
$\eta_{DES} = 46.0\%$ at all 8 intermediate sizes (5k–80k); omitted for brevity				
100,000	46.0	46.1	<0.1	51–66
With exception-based telemetry ($p_{exc} = 0.30$)				
10,000	14.0	13.8	<1.5	16–20
100,000	14.0	13.8	<1.5	16–20

[a] η_{DES} = DES-measured (30 MC replications, SD $< 0.001\%$). η_{analytic} = closed-form. Total channel utilization $\eta_{\text{total}} = \eta + 20.5\%$ baseline: 67% (stress), 26% (nominal).
[b] $\eta_{\text{eff}} = \eta_{\text{DES}}/\gamma$ for $\gamma \in [0.7, 0.9]$.

G. Topology Comparison and Baselines

Table XIV and Fig. 12 summarize scaling behavior.

Centralized baseline ($c = N/k_c$). Centralized processing does not diverge until $N \approx 10^6$ under $M/D/c$. This model captures compute-queue scalability only—not uplink scheduling or ground contact availability [9]. Cross-architecture overhead comparisons are restricted to hierarchical vs. sectorized mesh (both fully modeled at the communication layer).

Latency. Within-cycle queueing: ∼260 ms mean (Table XIII); AoI: 44–441 s (Table IX). Centralized round-trip is lower; the hierarchical advantage is fault tolerance.

Bandwidth–robustness tradeoff. The hierarchy saves bandwidth via $1/k_c$ aggregation but depends on coordinator availability (99.5%); the mesh provides local robustness (no coordinator dependency) at higher overhead with narrower scope (Table XV).

MAC contention. Both architectures are compared at $\gamma = 0.85$ (TDMA). The hierarchy achieves this via coordinator-assigned slots; the sectorized mesh would require distributed TDMA coordination (adding unmodeled overhead) or CSMA/CA ($\gamma \approx 0.3$–0.5, increasing mesh overhead by 1.7–2.8×).

H. Cluster, Duty Cycle, and Link Sensitivity

1) Cluster Size: Protocol overhead is nearly invariant to k_c ($\pm 0.1\%$; Table XVI). The trade-off is *latency*: smaller clusters increase regional queueing delay (at $k_c = 50$, 2,000 summaries/cycle vs. 1,000 at $k_c = 100$). $k_c = 100$–200 is favorable under synchronized scheduling.

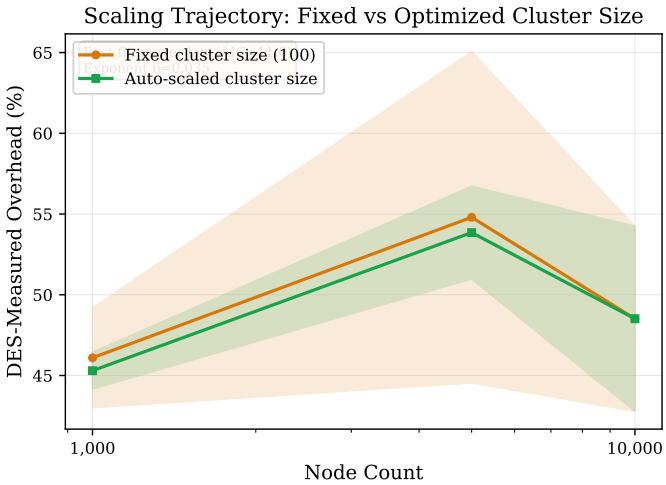

Fig. 10. Overhead trajectory (log-linear, DES-measured). Hierarchical $\eta \approx 46\%$ constant across 10^3–10^5. Dashed: exception telemetry.

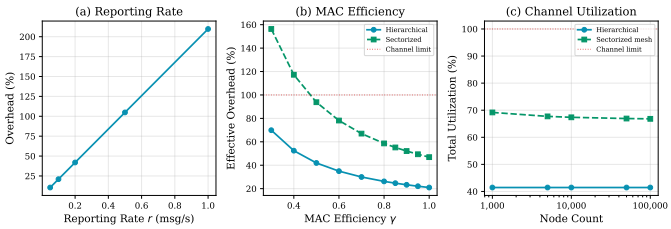

Fig. 11. Parametric sensitivity. (a) Overhead vs. reporting rate. (b) MAC efficiency γ vs. effective overhead. (c) Total channel utilization.

2) Coordinator Duty Cycle: The 24–48 h duty cycles occupy the Pareto frontier (Table XVII, Fig. 14). Power CV $\approx 12\%$ at $k_c = 100$; per-coordinator availability $> 99.99\%$ (two-state Markov, MTTF = 50 yr [24]). The 99.5% in Table XIV conservatively accounts for cascading effects.

3) Link Availability: Table XVIII summarizes coordination success under Bernoulli losses. Retransmission ($M_r = 2$) results are byte-budget delivery rates; TDMA airtime is not enforced. **Regime A** ($\geq$10 kbps or $k_c \leq 30$): retransmissions fit in the superframe. **Regime B** (RF-backup, 24–30 kbps, $k_c = 100$): effective $M_r = 0$ intra-cycle; recovery is inter-cycle (Section IV-C).

I. Packet-Level Validation

The message-layer models (analytical, DES, slot-sim) all implement the same sizing equations; their agreement is expected by construction. To provide genuinely independent validation, we construct a packet-level TDMA simulator that derives γ from CCSDS-standard framing rather than assuming it.

1) γ Decomposition from Physical-Layer First Principles: Each TDMA slot is decomposed into four sub-efficiencies (Table XIX):

$$\gamma_{\text{total}} = \gamma_{\text{framing}} \times \gamma_{\text{FEC}} \times \gamma_{\text{guard}} \times \gamma_{\text{acq}} \tag{13}$$

TABLE XIII
HIERARCHICAL LATENCY DECOMPOSITION ($N = 10^5$, $k_c = 100$)

Component	Value (ms)	Source
Member→coordinator propagation[a]	1.7	500 km/c
Coordinator processing (s_{proc})	5.0	Table IV
Within-cycle batch queueing (mean)[b]	250	$k_c \times s_{\text{proc}}/2$
Coordinator→regional propagation	1.7	500 km/c
Regional processing	2.0	Table IV
Total (mean)	**∼260**	
Total (P95)	**∼500**	$k_c \times s_{\text{proc}}$

[a]500 km intra-cluster diameter. [b]Batch queueing: all k_c reports arrive within one cycle; the coordinator processes them serially at 5 ms each. The last report waits $\sim k_c \times s_{\text{proc}} = 500$ ms. This is a $D[k_c]/D/1$ batch model, not steady-state $M/D/1$ (see Section III-B). Note: these latency values assume fluid-server processing and do not include the TDMA wait-for-slot delay (up to one full frame T_c); the slot-level simulator addresses superframe timing but not per-packet queueing latency.

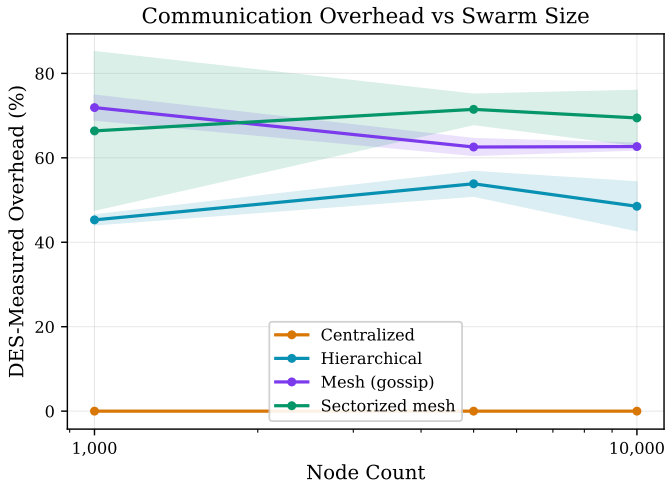

Fig. 12. Overhead vs. node count (log–log). Centralized ($c=1$) is *compute-queue only*—divergence reflects processing, not communications. Mesh saturates; hierarchical stays at ∼46%. Sectorized mesh provides different functional scope (local-neighbor monitoring vs. cluster-wide awareness); overhead values are not directly comparable to hierarchical.

TABLE XIV
COORDINATION TOPOLOGY COMPARISON

Architecture	Scalability Limit	O_{protocol}[a]	Failure Mode
Reference Bounds			
Centralized ($c=1$, bound)	∼10,000[b]	—[d]	Single point
Centralized ($c=N/k_c$)	∼10^6[c]	—[d]	Single point
Global-State Mesh (UB)	∼1,000	>100%	Distributed (99.99%)
Sectorized Mesh	$> 10^5$ (DES)	65–67%	Distributed
Architecture Under Study			
Hierarchical	10^5 (validated)	5–46%	Graceful (99.5%)

[a]Protocol overhead beyond the topology-invariant 20.5% baseline. Hierarchical range: ∼5% (exception telemetry, $p_{\text{exc}} = 0.10$) to ∼46% (stress-case full reporting).
[b]Processing limit at $c = 1$; propagation/spectrum limits persist. [c]Realistic provisioning ($c = N/k_c$, $\rho \leq 0.7$); binding constraint is uplink spectrum and latency, not processing. [d]Not applicable (compute-queue model only; no communication overhead modeled). Cross-architecture overhead comparison is valid only between the hierarchical architecture and comparators with equivalent functional scope. The sectorized mesh provides local-neighbor monitoring only.

TABLE XV
FUNCTIONAL CAPABILITY MATRIX

Function	Cent.	Hier.	Mesh	Global
Fleet command dissem.	✓	✓	—	—
Full cluster awareness	✓[a]	✓	—	✓
Local neighbor monitor	—	✓	✓	✓
Summary aggregation	✓[a]	✓	—	—
No coordinator depend.	—	—	✓	✓
Fits 1 kbps budget	—[b]	✓	✓	—

[a]At ground; not in-orbit autonomous. [b]Uplink-limited, not budget-limited.

The derived $\gamma = 0.760$ is below the $\gamma = 0.85$ assumed throughout this paper and below the $\gamma = 0.949$ from the slot-level model (Eq. 4), confirming that FEC and antenna acquisition collectively consume ∼19% of slot capacity.

2) Cross-Model Comparison: Table XX compares four verification tiers at equivalent parameters. The packet-level model tightens the feasibility envelope: at 24 kbps with CCSDS framing, ingress exceeds T_c (infeasible); at 30 kbps, margin shrinks to ∼900 ms but remains positive.

3) FEC Rate Sensitivity: The choice of FEC code rate has first-order impact on feasibility. At rate 1/2 LDPC, even 30 kbps is infeasible (ingress exceeds T_c). At rate 7/8, 30 kbps is the minimum viable design point. Without FEC (rate 1.0), 24 kbps becomes feasible—but only at the cost of increased BER. This trade-off allows practitioners to select the FEC rate appropriate to their link environment, with the sizing equations accepting the resulting γ.

4) Interpretation: The packet-level validation reveals that the feasibility boundary is tighter than the message-layer analysis indicates. At 24 kbps with CCSDS-standard framing (rate-7/8 LDPC, 5 ms acquisition dwell), the derived $\gamma = 0.760$ places the superframe in the infeasible region of Table VII. The recommended 30 kbps design point (Section IV-A) is confirmed as a physical necessity, not merely a design margin.

V. DISCUSSION

A. Validation Gap and Open Questions

Table XXI maps each key result to its verification status. The DES provides code verification (implementation consistency); the slot-level TDMA simulator provides model verification (superframe timing, pipeline decoupling, ARQ coupling). The packet-level TDMA simulator (Section IV-I) provides physical-layer validation by deriving γ from CCSDS-standard framing and confirming the feasibility boundary independently. Remaining validation—full MAC contention, antenna scheduling, multi-cluster shared-medium effects—requires NS-3 packet-level simulation.

B. Limitations

Physical-layer abstraction. MAC contention, link acquisition, and Earth-occlusion outages are not modeled; shared-medium contention would couple GE loss and coordinator ingress. **Priority queueing:** equal-priority messages; collision avoidance alerts should receive priority. **Failure models:** i.i.d.

TABLE XVI
HIERARCHICAL OVERHEAD AND LATENCY VS. CLUSTER SIZE[A]

Cluster	η (%)		Latency (ms)	
Size k_c	$N=10^4$	$N=10^5$	$N=10^4$	$N=10^5$
50	46.0	46.0	508	675
100	46.0	46.0	340	508
200	46.1	46.1	340	508
500	46.1	46.1	340	508

[a]Mean across 30 MC replications. Latency includes cycle-alignment ($T_c/2$) plus multi-tier propagation. Identical latency at $k_c \geq 100$ reflects the DES reporting granularity ($T_c = 10$ s cycle boundaries); differences below T_c are not resolved.

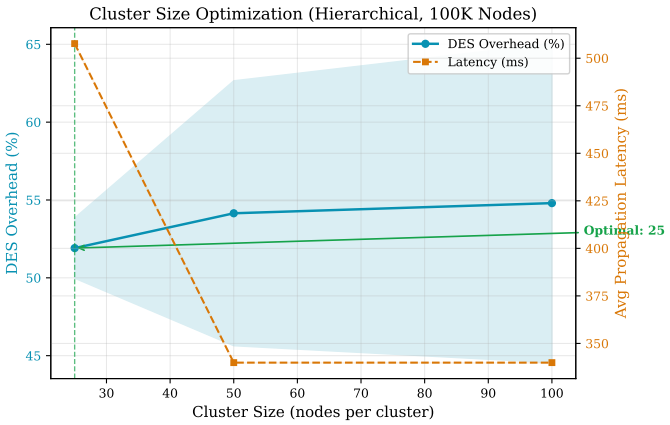

Fig. 13. Cluster size sensitivity: η invariant ($\pm 0.1\%$); latency drives the trade-off. $k_c = 100$–200 favorable.

TABLE XVII
COORDINATOR DUTY CYCLE TRADE-OFFS

Duty Cycle	Power Var.[a]	Handoff Success[b]	System Avail.[c]	Handoff Cost
1 h	<5%	95.0%	99.9%	High
6 h	8%	98.0%	99.8%	Moderate
24 h	12%	99.5%	99.5%	Low
48 h	18%	99.8%	99.2%	Very low
7 d	35%	99.9%	98.0%	Minimal

[a]CV of per-node power (derivation above). [b]Probability that state transfer (10–50 MB over optical ISL; reduced 25.6 kB payload over RF backup) completes. [c]Two-state Markov model with SWIM failure detection and Raft election.

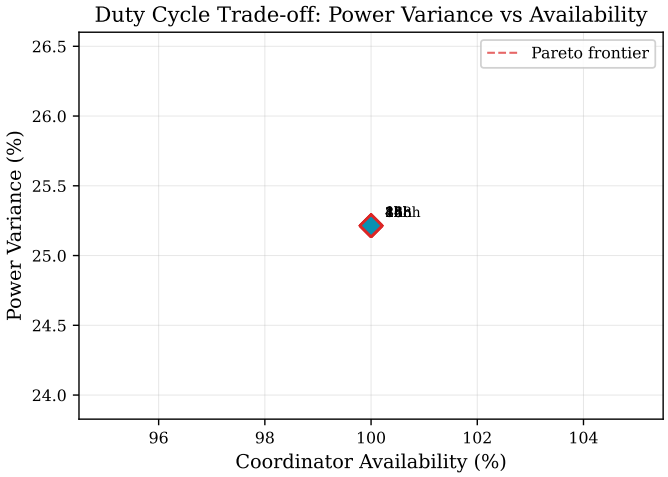

Fig. 14. Duty cycle trade-off in the power variance–system availability space. The 24 h and 48 h configurations offer the best balance.

exponential; correlated modes (SPE, batch defects) are future work.

Dynamic topology. An orbital mechanics analysis (J2 perturbation model for a Starlink-like Walker constellation: 53° inclination, 550 km, 72 planes, 22 sats/plane) quantifies re-association dynamics. Co-orbital clusters (nodes in the same plane) experience zero re-association: RAAN regression is identical for all nodes sharing altitude and inclination. Cross-plane encounters are infrequent: differential RAAN drift between adjacent planes is $\sim$0.069°/day, giving re-association periods of $\sim$5,200 orbits per boundary ($\sim$347 days). Fleet-wide: $\sim$0.014 events/orbit across 72 boundaries. Re-association overhead per event: one Raft election ($\sim$12.8 kB) + state handoff ($\sim$2 kB) = 14.8 kB; amortized overhead: <0.3% of byte budget. AoI transient during re-association: election (3 s) + state rebuild ($3 \times T_c = 30$ s) = 33 s, modest vs. P99 = 441 s (<8%). Cluster membership is fixed in the DES; a full Walker simulation with dynamic membership is planned as future work.

C. Message-Layer Design Equations Summary

For convenience, the key message-layer sizing relationships are collected here. All values assume message-layer operation; multiply bandwidth by $1/\gamma$ for MAC overhead. The γ parameter absorbs all physical-layer effects; Table VII provides breakpoint sensitivities for deployment-specific calibration.

- **Overhead:** $\eta = \eta_0 + \eta_{\text{cmd}}$; $\eta_0 \approx 5\%$; $\eta_{\text{cmd}} = p_{\text{cmd}} S_{\text{cmd}} \times 8/(C_{\text{node}} T_c)$.
- **Coordinator ingress:** $C_{\text{coord}} \geq k_c S_{\text{eph}} \times 8/T_c$ ($\geq$ 20.5 kbps at defaults).
- **AoI P99:** $T_c \lceil \ln(0.01)/\ln(1 - p_{\text{exc}}) \rceil$ (440 s at $p_{\text{exc}} = 0.10$).
- **GE bad-state recovery:** $p_{\text{success}} = 1 - p_B^{M_r+1}$ (27.1%); inter-cycle P95 = 4 cycles at $p_{BG} = 0.50$ (Fig. 8).
- **Safe-mode:** $\gamma_{\min} = \eta_{\text{total}}$; nominal survives Slotted ALOHA ($\gamma = 0.36$); stress requires $\gamma \geq 0.67$.
- **Fleet reuse:** $T_c^{\text{fleet}} = \max(T_c, \lceil f_{\text{RF}} N/(k_c F R) \rceil T_c)$; non-binding at $f_{\text{RF}} \leq 1.2\%$.

Uncertainty at MAC boundary. Physical-layer overheads enter through $1/\gamma$. Table VII shows first-order sensitivities; second-order interactions (contention $\times$ loss, antenna scheduling $\times$ half-duplex) require packet-level simulation.

VI. CONCLUSION

This study provides a message-layer sizing framework for hierarchical coordination in autonomous space swarms (10^3–10^5 nodes), decomposing feasibility into three layers: byte budget (η), MAC efficiency (γ), and TDMA airtime scheduling. The framework separates topology-dependent overhead (η_0) from workload-dependent command traffic (η_{cmd}) and characterizes correlated loss recovery via a Gilbert-Elliott Markov chain with literature-grounded parameterization [52].

Under the representative instantiation (Table IV): architecture-specific overhead $\eta_0 \approx 5\%$; stress-case $\eta_S \approx 46\%$

TABLE XVIII
COORDINATION SUCCESS VS. LINK AVAILABILITY ($N = 10^4$, $k_c = 100$). $M_r = 2$ COLUMNS APPLY TO **REGIME A ONLY** ($\geq$10 KBPS OR $k_c \leq 30$); FOR RF-BACKUP REGIME B (24 KBPS, $k_c = 100$), USE $M_r = 0$ COLUMNS.

p_{link}	**Msg. Delivery (%)**[b]		**Delivered η (%)**		**Offered**[a]
	$M_r = 0$	$M_r = 2$	$M_r = 0$	$M_r = 2$	$M_r = 2$
1.0	100.0	100.0	46.0	46.0	46.0
0.9	89.9	99.9	41.3	46.0	51.1
0.8	79.9	99.3	36.8	45.7	57.0
0.7	69.8	98.0	32.1	45.1	63.9
0.6	59.7	95.8	27.5	44.1	71.8
0.5	49.9	92.9	23.0	42.7	80.5[b]
0.4	39.9	89.0	18.4	40.9	90.2[b]

[a]Including retransmissions. [b]Per-message rate; per-cycle $\approx (1 - p_{\text{loss}})^{k_c}$.

TABLE XIX
γ DECOMPOSITION FROM CCSDS PROXIMITY-1 FRAMING ($S_{\text{EPH}} = 256$ B, 24 KBPS PHY)

Component	**Mechanism**	**Efficiency**	**Source**
Framing	ASM+addr+ctrl+FCS: 104 b OH	0.952	CCSDS 211.0-B-5 [28]
FEC	LDPC rate 7/8: 2152→2460 b	0.875	CCSDS 131.0-B-4 [56]
Guard	prop+turn+jitter: 4.7 ms	0.956	Prox-1 [57]
Acquisition	pointing dwell: 5 ms	0.955	Typical S-band ISL
Total derived		**0.760**	Product
Paper assumed		0.850	Conservative
Slot-level (Eq. 4)		0.949	No FEC/acq

TABLE XX
CROSS-MODEL COMPARISON ($k_c = 100$, NO LOSS)

Metric	**Analyt.**	**DES**	**Slot** ($\gamma = 0.85$)	**Pkt** ($\gamma = 0.76$)
Ingress 30 kbps (ms)	6,930	6,930	6,930	9,078
Margin 30 kbps (ms)	2,870	2,870	2,870	~900
Feasible @ 24 kbps	Y ($\gamma = 0.85$)	Y	Y	**N**
Feasible @ 30 kbps	Y	Y	Y	**Y**
ARQ infeas. (GE $M_r \geq 1$)	Markov	—	52.7% miss	Conf.

TABLE XXI
MODEL VALIDITY & CLAIM MAP

Result	**Analyt.**	**DES**	**Slot-sim**	**Pkt-Level**	**Unmod.**
$\eta_0 \approx 5\%$	§IV-E	<0.1%	—	—	—
$\eta_S \approx 46\%$	§IV-E	<0.1%	—	—	—
Coord. 24–30 kbps	Eq. 5	—	Exact	24 infeas.	—
Margin 623 ms	Tbl. VIII	—	Exact	—	—
γ sensitivity	Tbl. VII	—	Conf.	$\gamma = 0.76$	—
ARQ infeas. (GE)	Markov	—	52.7%	Conf.	—
1 kbps design	—	—	—	Link budget	—
AoI P99 $= 440$ s	Eq. 12	Conf.	—	—	—
GE P95 rec. 4 cyc.	Markov	Conf.	—	—	—
$T_{\text{cmd}}(G)$ stagger	Eq. 8	—	Conf.	—	—
TDMA decoupling	—	—	($M_r = 0$)	—	—
4-geometry	—	—		—	—
ARQ coupling	—	—	($M_r \geq 1$)	—	—
T_c^{fleet} reuse	Eq. 6	—	—	—	NS-3
Antenna sched.	—	—	—	—	Future
MAC contention	—	—	—	—	Future

with commands dominating; coordinator ingress 24–30 kbps at $\gamma = 0.85$; GE inter-cycle P95 recovery in 4 cycles; AoI P99 $= 440$ s under exception telemetry. Slot-level TDMA simulation confirms pipeline decoupling under $M_r = 0$ and quantifies ARQ coupling at $M_r \geq 1$. Fleet-level channel reuse is non-binding below $f_{\text{RF}} = F \cdot R \cdot k_c/N$ (Eq. 6).

Packet-level validation (Section IV-I) derives $\gamma = 0.76$ from CCSDS Proximity-1 framing with rate-7/8 LDPC, tightening the feasibility boundary: 24 kbps is infeasible with full physical-layer overhead; 30 kbps is the minimum viable design point. A link budget (Table VI) grounds the 1 kbps RF-backup constraint in UHF omnidirectional parameters (0.1 W, 400 MHz, 1000 km range). Under distributed command planning (3-round Raft consensus), the topology-invariant command overhead conclusion requires qualification: intra-cluster consensus traffic ($\sim$31%) competes with centralized commands (41%), making overhead workload-dependent.

The sizing equations generalize to any workload semantics, command model, and MAC efficiency. The four-tier verification taxonomy (Table XXI) now covers code verification, model verification, packet-level validation, and identifies the remaining NS-3 gap (full MAC contention and antenna scheduling).

DATA AVAILABILITY

Source code, configuration, MC datasets, TDMA slot-level simulator, packet-level TDMA simulator, and link budget calculator are at https://github.com/projectdyson/dyson (tag: `paper-02-v-bz`). Parameters: Table IV. Environment: Python 3.12, NumPy 2.x, SciPy 1.x, Matplotlib 3.x. Total MC wall-clock time: $\sim$90 min on commodity hardware.

ACKNOWLEDGMENT

An AI-assisted ideation exercise (Claude 4.6, Gemini 3 Pro, GPT-5.2; see [25]) motivated aspects of the coordinator architecture but is not validated here.

REFERENCES

[1] SpaceX, "Application for modification of authorization for the SpaceX Gen2 NGSO satellite system," FCC Filing SAT-MOD-20230320-00028, Mar. 2023. See also: J. McDowell, "Jonathan's Space Report," planet4589.org (non-archival; accessed Feb. 2026).

[2] T. Flohrer, H. Krag, and S. Lemmens, "Assessment of the current conjunction event prediction performance," in *Proc. 7th European Conf. Space Debris*, Darmstadt, Germany, 2017, pp. 1–8.

[3] Amazon, "Project Kuiper: An overview," 2023 (non-archival; accessed February 2026). [Online]. Available: https://www.aboutamazon.com/what-we-do/devices-services/project-kuiper

[4] C. McLaughlin, "OneWeb constellation management approach," in *Proc. 35th Space Symposium*, Colorado Springs, CO, 2019.

[5] G. K. O'Neill, *The High Frontier: Human Colonies in Space*. New York, NY, USA: Morrow, 1976.

[6] V. Badescu and R. B. Cathcart, Eds., *Macro-Engineering: A Challenge for the Future*. Springer, 2006.

[7] N. A. Lynch, *Distributed Algorithms*. San Francisco, CA, USA: Morgan Kaufmann, 1996.

[8] M. Brambilla, E. Ferrante, M. Birattari, and M. Dorigo, "Swarm robotics: A review from the swarm engineering perspective," *Swarm Intell.*, vol. 7, no. 1, pp. 1–41, Mar. 2013.

[9] J. R. Wertz, D. F. Everett, and J. J. Puschell, Eds., *Space Mission Engineering: The New SMAD*. Hawthorne, CA, USA: Microcosm Press, 2011.

[10] C. W. Reynolds, "Flocks, herds, and schools: A distributed behavioral model," in *Proc. 14th Annu. Conf. Comput. Graph. Interact. Tech. (SIGGRAPH)*, 1987, pp. 25–34.

[11] M. Dorigo, G. Theraulaz, and V. Trianni, "Swarm robotics: Past, present, and future," *Proc. IEEE*, vol. 109, no. 7, pp. 1152–1165, Jul. 2021.

[12] M. Dorigo and T. Stützle, *Ant Colony Optimization*. Cambridge, MA, USA: MIT Press, 2004.

[13] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. IEEE Int. Conf. Neural Netw.*, vol. 4, Perth, Australia, 1995, pp. 1942–1948.

[14] D. Karaboga and B. Basturk, "A powerful and efficient algorithm for numerical function optimization: Artificial bee colony (ABC) algorithm," *J. Global Optim.*, vol. 39, no. 3, pp. 459–471, 2007.

[15] E. Tolstaya, F. Gama, J. Paulos, G. Pappas, V. Kumar, and A. Ribeiro, "Learning decentralized controllers for robot swarms with graph neural networks," in *Proc. Conf. Robot Learn. (CoRL)*, 2020, pp. 671–682.

[16] J.-M. Lasry and P.-L. Lions, "Mean field games," *Japanese J. Math.*, vol. 2, no. 1, pp. 229–260, 2007.

[17] M. Huang, R. P. Malhamé, and P. E. Caines, "Large population stochastic dynamic games: Closed-loop McKean–Vlasov systems and the Nash certainty equivalence principle," *Commun. Inf. Syst.*, vol. 6, no. 3, pp. 221–252, 2006.

[18] Q. Li, F. Gama, A. Ribeiro, and A. Prorok, "Graph neural networks for decentralized multi-robot path planning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2020, pp. 11785–11792.

[19] DARPA, "OFFensive Swarm-Enabled Tactics (OFFSET)," 2022 (non-archival; accessed February 2026). [Online]. Available: https://www.darpa.mil/program/offensive-swarm-enabled-tactics

[20] U.S. Dept. of Defense, "Replicator initiative fact sheet," Aug. 2023 (non-archival; accessed February 2026). [Online]. Available: https://www.defense.gov/News/Releases/

[21] K. Maine, C. Devieux, and P. Swan, "Overview of IRIDIUM satellite network," in *Proc. WESCON*, 1995, pp. 483–490.

[22] L. Kleinrock, *Queueing Systems, Vol. I: Theory*. New York, NY, USA: Wiley, 1975.

[23] A. Demers, D. Greene, C. Hauser, W. Irish, J. Larson, S. Shenker, H. Sturgis, D. Swinehart, and D. Terry, "Epidemic algorithms for replicated database maintenance," in *Proc. 6th Annu. ACM Symp. Principles Distrib. Comput.*, 1987, pp. 1–12.

[24] J.-F. Castet and J. H. Saleh, "Satellite and satellite subsystems reliability: Statistical data analysis and modeling," *Reliab. Eng. Syst. Safety*, vol. 94, no. 11, pp. 1718–1728, 2009.

[25] Project Dyson Research Team, "Multi-model AI deliberation for engineering design exploration: Methodology and case studies," 2025. [Online]. Available: https://projectdyson.org/publications

[26] L. Lamport, "Time, clocks, and the ordering of events in a distributed system," *Commun. ACM*, vol. 21, no. 7, pp. 558–565, Jul. 1978.

[27] D. Ongaro and J. Ousterhout, "In search of an understandable consensus algorithm," in *Proc. USENIX Annu. Tech. Conf. (ATC)*, Philadelphia, PA, USA, 2014, pp. 305–319.

[28] Consultative Committee for Space Data Systems, "CCSDS 211.0-B-6: Proximity-1 space link protocol—data link layer," CCSDS, Blue Book, Sep. 2013.

[29] M. Handley, "Delay is not an option: Low latency routing in space," in *Proc. 17th ACM Workshop Hot Topics Netw. (HotNets)*, Redmond, WA, USA, 2018, pp. 85–91.

[30] DARPA, "Blackjack program," 2023 (non-archival; accessed February 2026). [Online]. Available: https://www.darpa.mil/program/blackjack

[31] A. Das, I. Gupta, and A. Motivala, "SWIM: Scalable weakly-consistent infection-style process group membership protocol," in *Proc. Int. Conf. Dependable Syst. Netw. (DSN)*, Washington, DC, USA, 2002, pp. 303–312.

[32] O. Brown and P. Eremenko, "Fractionated space architectures: A vision for responsive space," in *Proc. 4th Responsive Space Conf.*, Los Angeles, CA, USA, 2006, pp. RS4-2006-1002.

[33] A. Golkar and I. Lluch i Cruz, "The federated satellite systems paradigm: Concept and business case evaluation," *Acta Astronautica*, vol. 111, pp. 230–248, Jun.–Jul. 2015.

[34] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1520–1533, Sep. 2004.

[35] W. Ren and R. W. Beard, "Consensus seeking in multiagent systems under dynamically changing interaction topologies," *IEEE Trans. Autom. Control*, vol. 50, no. 5, pp. 655–661, May 2005.

[36] Consultative Committee for Space Data Systems, "CCSDS 734.2-B-1: Bundle protocol version 7," CCSDS, Blue Book, Sep. 2020.

[37] I. F. Akyildiz, E. Ekici, and G. Yue, "A distributed multicast routing scheme for multi-layered satellite IP networks," *Wireless Netw.*, vol. 9, no. 5, pp. 535–544, Sep. 2003.

[38] I. del Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low Earth orbit satellite constellation systems to provide global broadband," *Acta Astronautica*, vol. 159, pp. 123–135, Jun. 2019.

[39] V. Cerf, S. Burleigh, A. Hooke, L. Torgerson, R. Durst, K. Scott, K. Fall, and H. Weiss, "Delay-tolerant networking architecture," IETF, RFC 4838, Apr. 2007.

[40] ESA Space Debris Office, "ESA's annual space environment report," ESA, GEN-DB-LOG-00288-OPS-SD, 2024.

[41] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" in *Proc. IEEE INFOCOM*, Orlando, FL, USA, 2012, pp. 2731–2735.

[42] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 5, pp. 1183–1210, May 2021.

[43] I. Kadota, A. Sinha, E. Uysal-Biyikoglu, R. Singh, and E. Modiano, "Scheduling policies for minimizing age of information in broadcast wireless networks," *IEEE/ACM Trans. Netw.*, vol. 26, no. 6, pp. 2637–2650, Dec. 2018.

[44] D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed. Hawthorne, CA, USA: Microcosm Press, 2013.

[45] S. Alfano, "A numerical implementation of spherical object collision probability," *J. Astronaut. Sci.*, vol. 53, no. 1, pp. 103–109, Jan.–Mar. 2005.

[46] D. Bhattacherjee, W. Aqeel, I. N. Bozkurt, A. Aguber, B. Chandrasekaran, P. B. Godfrey, G. Laughlin, B. M. Maggs, and A. Singla, "Gearing up for the 21st century space race," in *Proc. 17th ACM Workshop Hot Topics Netw. (HotNets)*, Redmond, WA, USA, 2018, pp. 113–119.

[47] Consultative Committee for Space Data Systems, "CCSDS 133.0-B-2: Space packet protocol," CCSDS, Blue Book, Jun. 2020.

[48] S. Chien, J. Doubleday, K. L. Wagstaff, D. R. Thompson, and S. Schaffer, "Distributed spacecraft autonomy," in *Proc. IEEE Aerosp. Conf.*, Big Sky, MT, USA, 2024, pp. 1–10.

[49] J. A. Fraire, P. G. Madoery, and J. M. Finochietto, "Contact graph routing in DTN space networks: Overview, enhancements and performance," *IEEE Commun. Mag.*, vol. 59, no. 2, pp. 118–124, Feb. 2021.

[50] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-efficient communication protocol for wireless microsensor networks," in *Proc. 33rd Hawaii Int. Conf. Syst. Sci.*, Maui, HI, USA, 2000, pp. 1–10.

[51] IEEE, "IEEE Std 1012-2016: Standard for System, Software, and Hardware Verification and Validation," IEEE, New York, NY, USA, 2017.

[52] E. Lutz, D. Cygan, M. Dippold, F. Dolainsky, and W. Papke, "The land mobile satellite communication channel—recording, statistics, and channel model," *IEEE Trans. Veh. Technol.*, vol. 40, no. 2, pp. 375–386, May 1991.

[53] ITU-R, "Recommendation ITU-R P.681-11: Propagation data required for the design of Earth-space land mobile telecommunication systems," Geneva, 2019.

[54] J.-Y. Le Boudec and P. Thiran, *Network Calculus: A Theory of Deterministic Queuing Systems for the Internet*. Berlin, Germany: Springer, 2001.

[55] Consultative Committee for Space Data Systems, "CCSDS 132.0-B-3: TM space data link protocol," CCSDS, Blue Book, Sep. 2020.

[56] Consultative Committee for Space Data Systems, "CCSDS 131.0-B-4: TM synchronization and channel coding," CCSDS, Blue Book, Sep. 2020.